Lecture-31
External Flows IV: Pressure Gradients in a boundary layer flow

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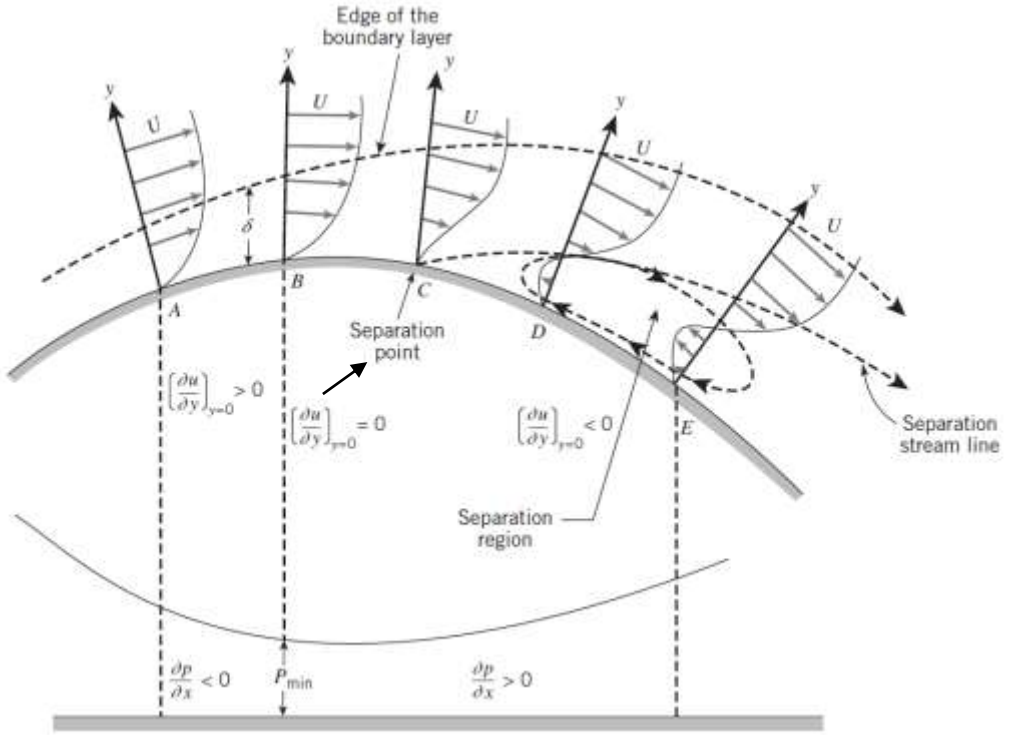
Learning Objectives

By the end of this lecture, you should be able to

1. Differentiate between **favorable**, **zero**, and **adverse pressure gradients** based on freestream velocity variation.
2. Explain why an **inflection point in the velocity profile** may develop under **adverse conditions**
3. Explain why **turbulent boundary layers resist adverse pressure gradients** and **delay separation** significantly better than laminar boundary layers

Pressure gradients in boundary layers

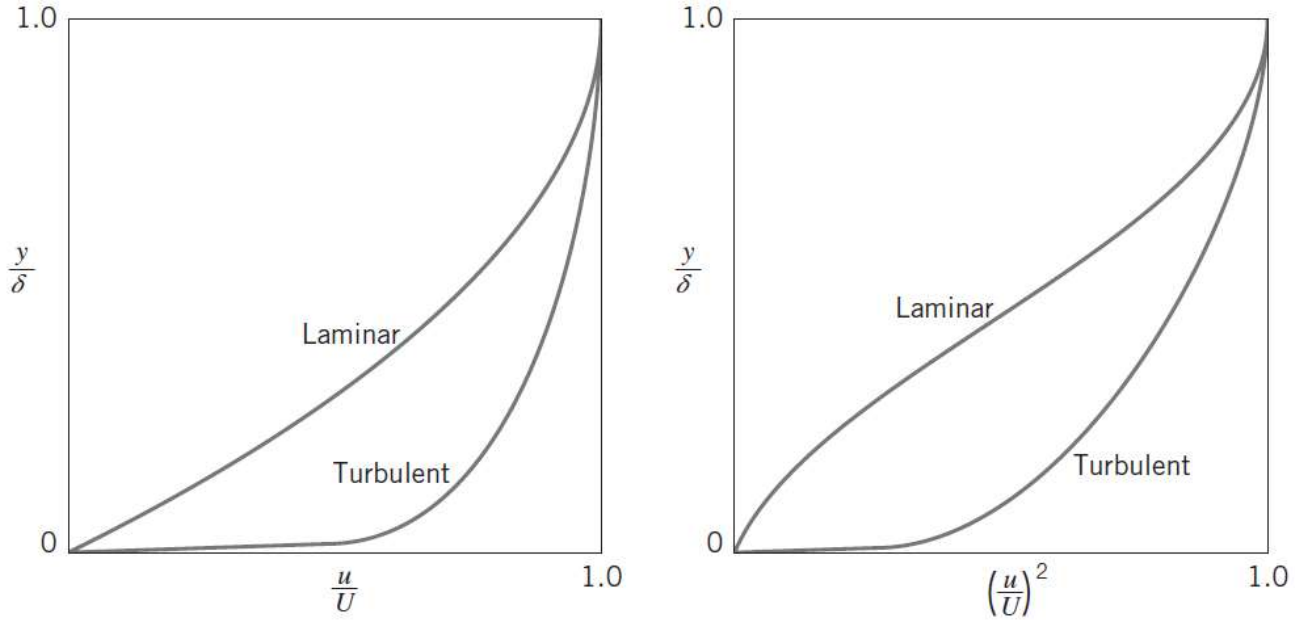
- In the region from A to B, **pressure decreases** and **flow accelerates**. This is favorable for BL formation.
- At point B, the pressure gradient is zero.
- At point C, the **pressure gradient is sufficiently positive** for BL separation. This is adverse for BL formation.



Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Flow separation in Laminar vs Turbulent BL

- The turbulent boundary layer profile is **more blunt** than the laminar case.
- Because of the **larger momentum flux**, a turbulent boundary layer is able to **better resist flow separation**.



(*a*) Velocity profiles (*b*) Momentum-flux profiles

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Revisiting the learning objectives

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